

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Optimized AI and HPC application performance on RISC-V silicon with vector extensions, identifying and resolving CPU and memory hierarchy bottlenecks.
- Developed Python scripts to automate performance benchmarking and instrumentation at the operating system level, integrating with CI/CD pipelines for consistent testing.
- Contributed to the design and implementation of low-level C++ code for performance-critical libraries, focusing on efficient use of CPU caches and memory bandwidth.
- Analyzed workload characterization data from simulators and emulators (e.g., Qemu, Gem5) to decompose application execution on underlying hardware.
- Collaborated with hardware teams to influence RISC-V architecture design by providing feedback on software performance implications.
- Debugged and troubleshooted complex performance issues across CPU, memory, and power utilization metrics on pre-silicon and silicon environments.
- Participated in regular code reviews and design document discussions with cross-functional teams to ensure technical consensus and unblock issues.

SKILLS

Core Technologies – C, C++, Rust, Python, Assembly Language, Fortran, SQL

Performance Tools – Linux perf, flamegraphs, sysbench, lmbench, iperf, Intel VTUNE, ARM SPE

Architecture & OS – Computer Architecture, CPU, Cache, Memory Hierarchy, Buses, Operating Systems, Linux

AI/HPC Benchmarks – SPEC CPU, MLPERF, BERT, ResNet, Graph500, HPGMG, CoMD, Linpack, CoreMark

Development Practices – Git, CI/CD, Jira, Docker, Kubernetes, Emulators (Qemu, Gem5, Spike)

System Level – Storage, Power Utilization, Firmware, Compilers, Middleware, Embedded Systems, Microcontrollers

PROJECTS

RISC-V Vector Extension Performance Tuning

- Tuned a benchmark suite (e.g., SPEC ACCEL) for RISC-V processors, achieving a **15% performance uplift** by optimizing vector instruction usage.
- Developed custom Python scripts to analyze low-level performance counters and generate detailed reports on cache miss rates and branch prediction accuracy.
- Integrated performance profiling into a Jenkins CI/CD pipeline to automatically track regressions and report on key performance indicators.

HPC Application Bottleneck Analysis (MSc Project)

- Analyzed the performance of a parallel HPC application (e.g., a molecular dynamics simulation) using C++ and profiling tools like `perf` and `gprof`.
- Identified critical bottlenecks related to memory bandwidth and inter-process communication, implementing targeted optimizations that reduced execution time by **22%** on a multi-core system.
- Documented findings and optimizations in a technical report, including a comparison of different optimization strategies and their impact on scalability.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)